Yukang Shen

shenyukang99@gmail.com |Homepage

Research Interests: ***Embodied AI, World Models, Spatial Perception, Spatial Reasoning, Safe and Reliable AI Systems***

OBJECTIVE

Interested in developing reliable perception and reasoning systems for embodied AI in real-world environments. My research focuses on integrating spatial perception, world models, and safety-aware decision mechanisms for medical and assistive AI applications.

EDUCATION

Kennesaw State University, M.S. Software Engineering **2025.1 – Present**
Graduate Research Assistant

Heilongjiang University, B.Eng. Computer Network Engineering **2016.9 – 2020.7**
National Encouragement Scholarship; First-Class Scholarship (2018, 2019)

RESEARCH EXPERIENCE

Synthetic-First Data Engine for Medical Perception (Paper in prep) Project Homepage

- Designed and implemented a **hierarchical perception prototype** for deltoid injection guidance, integrating **synthetic data generation**, **segmentation**, **geometric keypoint extraction**, and **safety-zone reasoning**.
- **Developed dataset curation tools** and multi-view generation templates, along with pseudo-label refinement modules to improve anatomical boundary precision.
- Investigating **synthetic-to-real transfer** using segmentation metrics, boundary distance, and medically grounded safety indicators.

Safety-Aware Embodied AI for Real-World Task Assistance (Research Direction)

- Investigating perception pipelines that connect segmentation, geometric reasoning, and spatial understanding for embodied decision-making.
- Exploring **safety-aware perception and reasoning strategies** for deployable medical assistant agents.
- Investigating **agent-centric learning frameworks** that integrate data generation, reinforcement learning, and human-in-the-loop feedback for continual adaptation in embodied systems.

HONORS & AWARDS

Second Place, KSU C-Day Computing Showcase (Fall 2025).

WORK EXPERIENCE

City University of Hong Kong, Research Assistant **2024.11 – 2025.1**
Tech: Python, React

- Developed a secure, encrypted data-marketing platform supporting cross-institution research collaboration.
- Implemented frontend dashboards and backend API modules focused on privacy, data integrity, and workflow usability.

SenseTime Group Limited, Software Engineer **2020.11 – 2023.9**
Tech: TypeScript, React, Electron, NodeJS, Webpack, CI/CD

- Awarded **CIT Team Best Employee (2021)** for outstanding engineering impact and delivery.
- Led frontend development across shared component libraries, internal development tooling, and enterprise platforms.
- Built and maintained group-wide OA systems and Enterprise WeChat automation modules.
- Developed AR navigation and enterprise platforms used across multiple business units.

PUBLICATIONS

6G-enabled Edge AI for Metaverse: Challenges, Methods, and Future Research Directions. JCN 2022. **Authors:** Luyi Chang, Zhe Zhang, Pei Li, Shan Xi, Wei Guo, **Yukang Shen**, Zehui Xiong, Jiawen Kang, Dusit Niyato, Xiuquan Qiao, Yi Wu.

SKILLS

Programming: Python, PyTorch, TypeScript, Node.js, React, Electron,
AI: Computer Vision, 3D Vision, SLAM, Multimodal Perception **Systems:** Synthetic Data Pipelines, Dataset Tooling